

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

26zfAn2dw6gNzb2yNVECGLBxgGxPG2qJdY3pKRTk2oX1

Date: 5/1/2025

Compounds: Caffeine, Diphenhydramine, Famotidine

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Cafemotihydramine

Formula: C28H38N6O3

Weight: 510.63

Drug-likeness: 0.62

Activity:

Potential antihistamine with gastric acid suppression and CNS stimulatory effects

Target Analysis

Histamine H1 receptor

Binding Affinity: -8

Reduction in allergic reactions

Histamine H2 receptor

Binding Affinity: -9.5

Decreases in gastric acid secretion

Adenosine A1 receptor

Binding Affinity: -7.5

Increased wakefulness and CNS activity

Toxicity Profile

Risk Level: Moderate

Affected Systems: CNS, Heart, Digestive system

Mitigation Suggestions:

- Limit dosage
- Monitor heart function
- Avoid use in individuals with pre-existing cardiac conditions

Pharmacokinetics

Absorption:

Rapid absorption due to lipophilic nature

Distribution:

Wide distribution in the body, can cross the blood-brain barrier

Metabolism:

Primarily hepatic via cytochrome P450 enzymes

Excretion:

Renal excretion of metabolites

Half-life: Estimated to be around 8-12 hours

Detailed Analysis

Cafemotihydramine represents a novel attempt to combine the therapeutic activities of caffeine, diphenhydramine, and famotidine into a single compound. By integrating their chemical frameworks, we aim to harness the antihistamine capabilities of diphenhydramine, the gastric acid suppression effect of famotidine, and the CNS stimulating effect of caffeine. The synthesis involves the strategic merging of structural components to retain their respective active sites, resulting in a compound that could potentially improve patient adherence by reducing the pill burden in those requiring multiple therapies for allergies, gastric upsets, and to combat drowsiness related to antihistamine use. Through computational modeling, Cafemotihydramine was found to have moderate drug-likeness and acceptable pharmacokinetic properties, although careful monitoring for CNS overstimulation and cardiac issues is advised. Initial research supports the design of Cafemotihydramine as a multifunctional agent, prompting further exploration into its clinical applications.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>